

ESSAY ON VOLATILITY CLUSTERS AND TIME SERIES PREDICTION

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Abstract—Volatility moves in financial markets are rare and sporadic i.e., periods of low volatility typically follow each other until the regime change due to technical and/or fundamental exogenous factors. Subsequently higher volatility tends to lead to even higher volatility and vice versa. Such dynamic systems require alternative ways of representation than a static multi-layer perceptron (MLP). The prevalent models assume L-stable distributions with independent and stable increments. This contrasts with what is observed in real world. L-stable distributions miss one of the main features of financial markets – the alternation of periods of large price changes with periods of small price changes. To correct for this deficiency another, self-affine process had been introduced – Fractional Brownian Motion (FBM). FBM does not capture fat tails or fluctuations in volatility that are unrelated to the predictability of future returns. In summary, both models have strong scale-invariance property, in which the distribution of returns over different sampling intervals are identical. This property is clearly at odds with empirical observations.

The key idea in this paper is that Generative Adversarial Networks (GAN) combined with fractality of data is better suited to manage volatility time series between different shock events because it is structured to maintain a memory of older points in the time series and continuously learn from them. Episodic memory that is used in GANs maintains explicit record of past events. In order to make decision, the action is chosen that has the highest value based on the outcomes of past similar situations. There are other, older models such as time-delayed window input vector (TDNN nets) and recurrent structures such as ‘Jordan’ and ‘Elman’ but in this study we focus on a more modern way of looking at dynamic time series through semi-supervised framework.

Keywords: K-Means, FCM, IFCM, Intuitionistic fuzzy sets, Volatility, Neural Basis Expansion, NBEATS, Generative Adversarial Network, GAN.

I. INTRODUCTION

Measuring and forecasting volatility has received a lot of attention from both the academia and market practitioners ever since Chicago Board of Trade introduced VIX index in 1993. In 2003, Goldman Sachs together with CBOE came up with a novel way to measure expected volatility which has been widely used ever since by academia, traders and risk managers.

When volatility is interpreted as uncertainty it becomes a key input in portfolio investment decisions when pricing derivative instruments and in particular options. In 1996 Basel

Accord also started requiring modelling and forecasting of volatility as a compulsory input to risk management for financial institutions around the world. Ruipeng Liu and Rangan Gupta studied the role of investors’ uncertainty as measured by the conditional volatility of investor sentiment in forecasting volatility of six major stock markets using bivariate Markov-switching multifractal model (MSM). Their results demonstrated that forecasting stock market volatility in combination with information associated with investors’ uncertainty generates smaller forecasting errors relative to univariate MSM models which does not include the metric of uncertainty. The superior performance of the bivariate MSM models highlights the need to capture long memory and structural breaks when forecasting stock market volatility based on information contained in investors’ uncertainty.

Furthermore, since financial market volatility is affected by investors’ uncertainty, policymakers must realise that the effect of such uncertainty is likely to prolong the negative impact on the economy through the negative feedback loop on real macroeconomic variables and delays in investment and consumption decisions. When there is a rise in historical volatility, a stock’s price will also move more than normal. At this time, there is an expectation that something has changed. If the historical volatility is dropping, on the other hand, it means any uncertainty has been reduced significantly, so things return to the way they were.

In 2012, CBOE introduced the Volatility of Volatility Index or VIX, also referred to as vol of vols and this is what this paper is focusing on. VVIX is calculated in the same way as VIX with one exception that the inputs are time series of VIX instead of S&P 500 options. VVIX measures the expected volatility of VIX. The formula for the VIX calculation [5] is:

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2 \quad (1)$$

where:

- $VIX = \sigma * 100$,
- T = Time to expiration,
- F = Forward index level derived from index option prices,
- K_0 = First strike below the forward index level, F

- K_i = Strike price of i th out of the money option; a call if $K_i > K_0$ and a put if $K_i < K_0$; both put and call if $K_i = K_0$.

$$\delta K_i = \frac{K_{i+1} - K_{i-1}}{2}$$

- R = Risk-free interest rate to expiration
- $Q(K_i)$ = The midpoint of the bid-ask spread for each option with strike K_i .

Academic papers focusing on various forecasting models including GARCH, Monte Carlo, Support Vector Machines, LSTM, Random Forest and other more up to date techniques. Most of these academic studies focus on forecasting VIX (volatility) time series. There has been a relative scarcity of work and the author has not found any that focus on VVIX or vol of vols utilising modern approaches such as fractals or deep reinforcement learning and GANs. Some work has been done on forecasting VVIX using Bayesian analysis and Monte Carlo Markov chain.

Volatility is a critical input to numerous financial market operations and it becomes a tool to assess the risk of exposure for portfolio managers, investors and others [4, 6–9]. The modelling of volatility is a synthetic task, because it cannot be observed directly. Indeed, it can only be measured by looking at the extent of the option prices movement mathematically. Autoregressive moving average (ARMA) process, can not capture volatility directly.

A. CLUSTERING VOLATILITY WITH FS AND IFS

Big data clustering applications need to deal with feature values subject to uncertainty, thus the distances that determine the membership of an object to a cluster will also be subject to uncertainty. Therefore, the possibility of erroneous membership levels in the cluster's formation is possible.

Available fuzzy clustering approaches do not assume any information about ambiguity at the feature level. This paper considers the challenge to deal with such kind of information and paves some thoughts based on [6-8], [24] about a generalisation to the FCM.

B. RESULTS AND DISCUSSION

We examined the daily VIX dataset from January 1990 to April 2022, provided by Bloomberg Data services. Data from this period is preferable and less noisy with good predictive value.

VIX is a weighted average of implied volatilities as measured from the call and put prices. It measures 30-day expectation of volatility given by a weighted portfolio on the S&P 500 Index. Periods of low volatility typically follow each other until the regime change and subsequently higher volatility tends to lead to even higher volatility and vice versa.

We apply hard and soft clustering techniques on the VIX. K-means clustering, divides a set of n observations into k clusters comprised of observations with similar characteristics.

To find the number of clusters we use the 'Elbow' method and 'Silhouetter' analysis, on normalised data, using 'StandardScaler' technique.

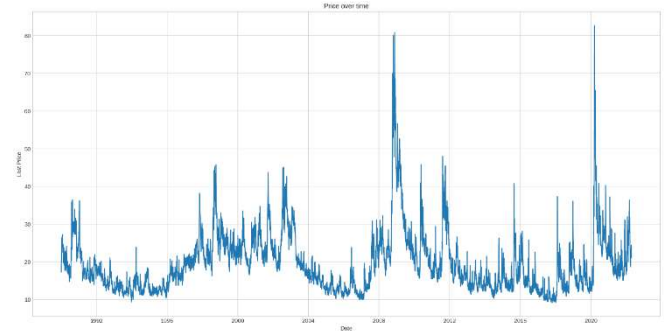


Figure 1. VIX Reporting period 1990 to 2022.

II. K-MEANS

Visually, we are looking for an elbow of this curve. It appears in the region of 2–4 clusters.

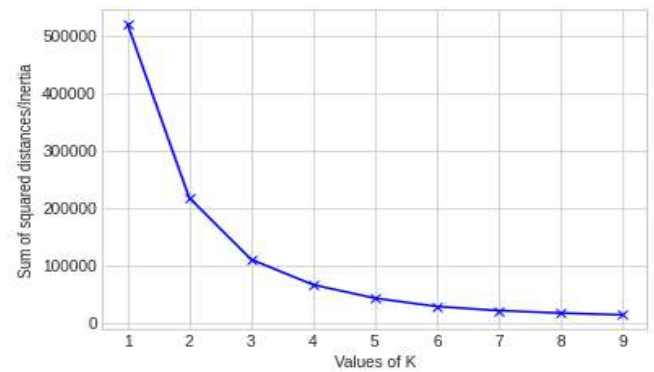


Figure 2. Optimal K value

Elbow method uses the concept of WCSS value. WCSS stands for Within Cluster Sum of Squares, which defines the total variations within a cluster. Visually, we are looking for the sharp bend, pointing to an optimal range of 2–4 clusters.

The 'Silhouette' method measures the cohesion of a point is similar within its own cluster as opposed to other clusters.

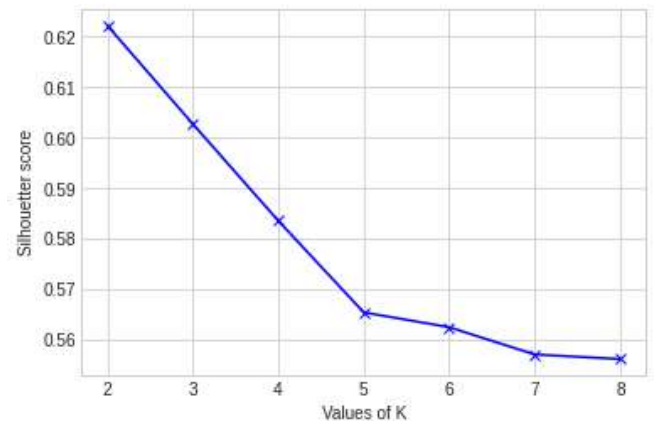


Figure 3. Silhouette for optimal K value

We produce an output with two and five clusters (Figure 4). The top graph shows two clusters identified as horizontal lines and the bottom graph five clusters

The two clusters graph simply groups the data into intervals of low and high volatility. The output shows when

the volatility is low it tends to stay low and when it goes high it tends to remain high.

The five cluster output splits the data into periods of very low volatility, low volatility, medium volatility, high volatility, and extremely high volatility.

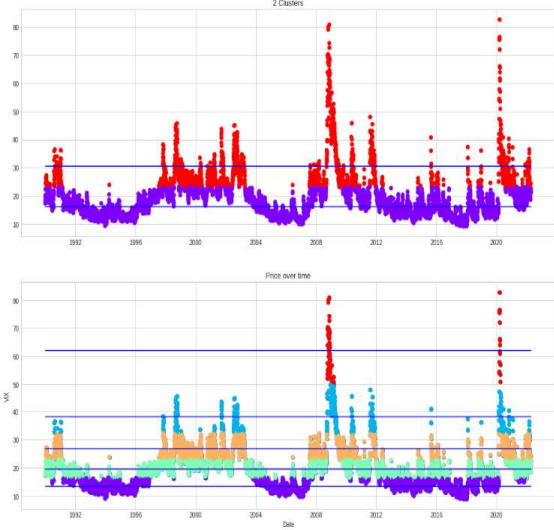


Figure 4. Periods of Volatility-Two Vs Five

We observe a silhouette score of 0.554. The observed silhouette score falls within the accepted range $[-1, 1]$. The silhouette score of 0.554 means that the clusters are reasonably dense and separated from each other.

III. FUZZY C-MEANS (FCM)

In FCM algorithm, an object can belong to more than one cluster. Plotting the Fuzzy C-Means periods 0–4 gives us the following results (Figure 5).

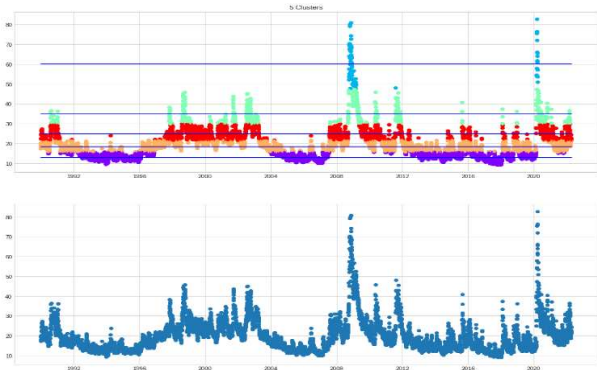


Figure 5. Periods of Volatility and Fuzzy C-Means

Based on FCM, data are splitted into periods of very low volatility, low volatility, medium volatility, high volatility, and extremely high volatility.

We observe a silhouette score of 0.555 possibly due to random state initialisation. The silhouette score of 0.555 means that the clusters are reasonably dense and separated from each other.

IV. INTUITIONISTIC FUZZY SETS AND IFCM UNITS

A fuzzy set non-membership value is calculated as a complement of the membership value to 1. However, in reality because of uncertainty, the non-membership is not always

equal to one minus the membership value. To deal with this uncertainty, Atanassov proposed another higher order fuzzy set called IFS [1–3]. An IFS $\tilde{A}$ in X is given by:

$$\tilde{A} \cong \{ \langle x, \mu_{\tilde{A}}(x), \nu_{\tilde{A}}(x) \rangle | x \in X \}, \quad (2)$$

where X is a universe of discourse and $\mu_{\tilde{A}}(x) : X \rightarrow [0, 1]$, $\nu_{\tilde{A}}(x) : X \rightarrow [0, 1]$ with the condition

$0 \leq \mu_{\tilde{A}}(x) + \nu_{\tilde{A}}(x) \leq 1; \forall x \in X$ and $\mu_{\tilde{A}}(x)$, $\nu_{\tilde{A}}(x)$ denote membership and non membership degree, respectively.

For each IFS $\tilde{A}$ in X , the hesitation degree should be considered. The hesitation degree of an element $x \in X$ is defined as:

$$\pi_{\tilde{A}}(x) = 1 - \mu_{\tilde{A}}(x) - \nu_{\tilde{A}}(x), \quad (3)$$

where $\pi_{\tilde{A}}(x)$ is hesitation degree and should satisfy the elementary condition of intuitionism, i.e.,

$0 \leq \pi_{\tilde{A}}(x) = 1 - \mu_{\tilde{A}}(x) - \nu_{\tilde{A}}(x) \leq 1$. In the literature, two fuzzy complements or IFS generators are used to construct intuitionistic fuzzy set: Sugeno's and Yager's [15]. The fuzzy complement function is defined as:

$$N(\mu(x)) = g^{-1}(g(1) - g(\mu(x))), \quad (4)$$

where $g(\cdot)$ is an increasing function and $g : [0, 1] \rightarrow [0, 1]$.

Yager's class can be generated by using the following function:

$$g(x) = x\lambda, \quad (5)$$

Non-membership values are calculated from Yager's intuitionistic fuzzy complement $N(x)$. The IFSs using Yager's intuitionistic fuzzy complement become

$$A_{\lambda}^{IFS} = \{ \langle x, \mu_A(x), 1 - \mu_A(x)^{\frac{1}{\lambda}} \rangle | x \in X \}, \quad (6)$$

Sugeno's negation can be generated using the following function:

$$g(x) = \frac{1-x}{1+\lambda x}, \quad (7)$$

Non-membership values are calculated from Sugeno's intuitionistic fuzzy complement $N(x)$. IFS constructed using Sugeno's fuzzy complement is as follows:

$$A_{\lambda}^{IFS} = \{ \langle x, \mu_A(x), \frac{1-\mu_A(x)}{1+\lambda\mu_A(x)} \rangle | x \in X \}, \quad (8)$$

In soft clustering methods, the membership value is computed based on a distance function [14–20]. The authors are currently engaged in creating a Python code that can implement IFS in Python and apply it to volatility clustering. The next step will be to replace Yager and Sugeno functions with a different power law function and fuzzify the Hurst Exponent in the power law function. The remaining part to be completed is to merge the FBM with IFS to create a new process which we called Intuitionistic Fuzzy Fractal Brownian motion.

V. NEURAL BASIS EXPANSION (NBEATS)

Oreshkin et al., [10] proposed a novel architecture for univariate time series forecasting. They applied it to a variety of TS forecasting problems using non-overlapping competition datasets: M4, M3 and Tourism. The results

significantly outperformed traditional econometric forecasting techniques.

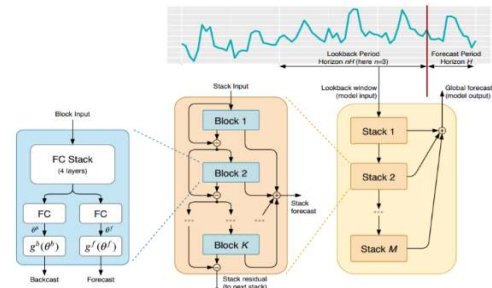
NBEATS is analysis for interpretable time series forecasting. The focus is on solving the univariate times series point forecasting problem using deep learning architecture based on backward and forward residual links and a very deep stack of fully connected layers. The architecture has several desirable properties, being interpretable, applicable without modification to a wide array of target domains, and fast to train.

The model consists of a sequence of stacks, each of which combines multiple blocks. The blocks connect feedforward networks via forecast and backcast links. A block “removes the portion of the signal ... it can approximate well” [9]. Then the block sets its focus on the residual error, which the preceding blocks could not disentangle. Each block generates a partial forecast, with its focus set on the local characteristics of the time series. The stack aggregates partial predictions on the blocks it consists of, and then passes the results on to the next stack. The purpose of the stack is to identify non-local patterns along the complete time axis by “looking back.” Finally, the partial forecasts are aggregated into an overall forecast at the model level. N-BEATS takes as its hyperparameters:

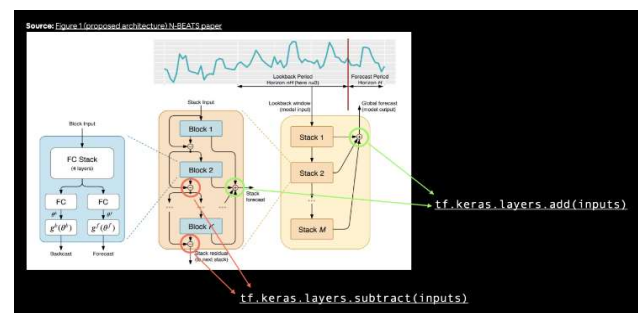
- i. The size of the input and output layers (constants INLEN and N_FC) must be sufficient to assign a node for each feature in the source data. The length of the input segment should not be less than the order of seasonality, otherwise the learning process will have more difficulty combining the segments. For efficient memory usage, set them to a power of 2.
- ii. The number of blocks in a stack (BLOCKS).
- iii. The width of each fully connected layer in each block of a stack: its node count (LWIDTH).

The batch size determines the number of cases the model will process before updating its matrix weights. To effectively align it with the memory structure of your system, set it to a power of 2. Very large batch sizes can skew gradients down only in one direction, and the model can get stuck at a sub-optimal minimum. Smaller batch sizes will cause the gradient descent to bounce around in different directions and can result in lower accuracy, but they also tend to prevent the model from overfitting. The most frequent recommendation is to choose an initial batch size of 32. Since our dataset has a frequency of 24 hours per day, we set the batch size to the following binary limit that can handle 24 time steps: 32. The epochs tell the model the number of cycles the exercise it has to perform. In each epoch, the model processes the entire training set, performing one forward pass and one back pass. Allowing for some oversimplification, the product of these hyperparameters defines the tensor size of the model. Large parameter values can cause it to reach the system's memory limit and cause exponentially longer processing times. While small parameter values may not be sufficient to reflect complex patterns in the source data. We obtain a probabilistic forecast using quantile regression. This is an option we can exercise in all deep forecasting models. The loss function of a neural network can be constructed using quantile loss function. Then quantile regression will not only compute a central forecast value at each time step, a point estimate, but will draw uncertainty bands about it. Pairs of quantiles like 1%/99% or 10%/90% express the range over which the forecast value can vary, above or below the central value. In

future work, we are planning to use transformers which is a relatively new concept in neural networks. Unlike univariate time series, transformer will learn to integrate various factors that influence VIX as regressors and use multivariate time series with multiple time frames. For example, we could split the time intervals into minutes, hours and days. This would provide a complex time series for a neural network to learn from.



Source: <https://medium.com/@jonathanbechtel/kerasbeats-an-easy-way-to-use-n-beats-in-keras-395b24c5cc28>

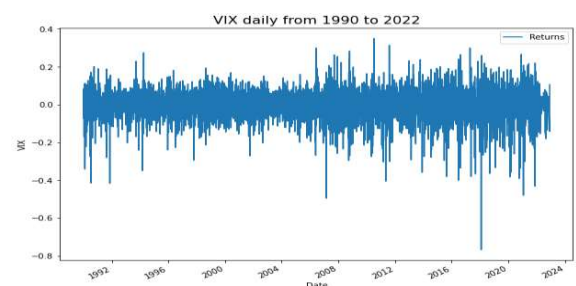


(Boris N. Oreshkin, 24 May 2019)

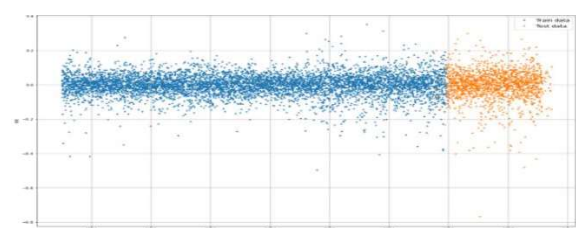
Next, we will detrend data and remove seasonality. We will compare NBEATS against naïve model and some other Deep learning models.

4. RESULTS

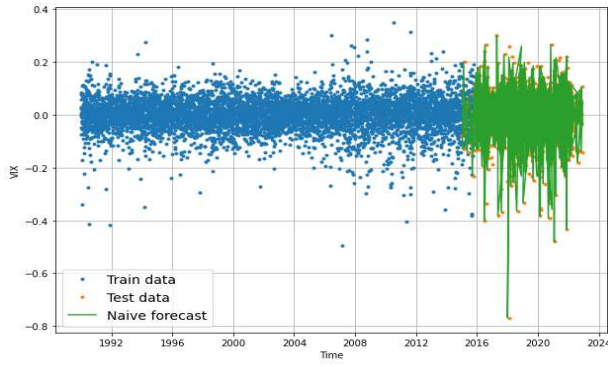
As was already mentioned, we ran data from January 1990 to April 2022. Graph below shows scaled log returns of VIX data series.



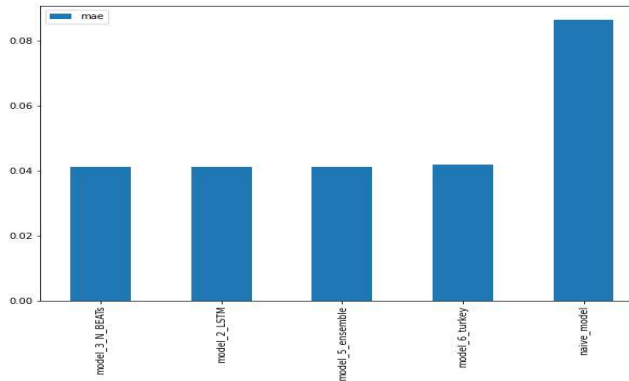
Train and test split



Naïve forecast



NBEATS model seems to do quite well compared to naïve model, but it is not a lot better than other deep learning models such as Conv1d and LSTM.



	mae	mse	rmse	mape	mase	
naive_model	0.086481	0.015316	0.086481	421354.18	0.999	
model_2_LSTM	0.041248	0.003253	0.057037	59694.81	0.672	
model_3_N_BEATs	0.041183	0.003258	0.057076	56545.72	0.671	
model_5_ensemble	0.041321	0.003274	0.057219	46762.38	0.673	
model_6_turkey	0.041824	0.003423	0.042492	33452.46	0.681	

VI. A GENERATIVE ADVERSARIAL NETWORK (GAN)

A generative adversarial network (GAN) [9] is a class of machine learning frameworks designed by Ian Goodfellow and his colleagues in June 2014. Two neural networks compete in a zero-sum game, where one agent's gain is another agent's loss.

With a training set, this technique learns to generate new data with the same statistics as the training set. For example, an image-trained GAN can produce new images that appear realistic at least to a human observer, with many realistic features. Though originally proposed as a form of generative model for unsupervised learning, GANs have also proved useful for semi-supervised learning, fully supervised learning, and reinforcement learning.

The core idea of a GAN is based on the "indirect" training through the discriminator, another neural network that can tell how much an input is "realistic", which itself is also being updated dynamically [4]. This basically means that the generator is not trained to minimize the distance to a particular frame, but rather to fool the discriminator. This enables the model to learn in an unsupervised manner.

GANs are like mimicry in evolutionary biology, with an evolutionary arms race between both networks. To learn the generator's distribution $p(g)$ over data x , we define a prior on input noise variables $p(z)$ then represent a mapping to data

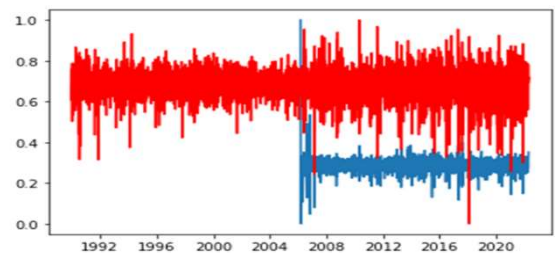
space as $G(z; \Theta_g)$, where G is a differentiable function represented by a multilayer perceptron with parameters Θ_g . We also define a second multilayer perceptron $D(x; \Theta_d)$ that outputs a single scalar. $D(x)$ represents probability that x came from the data rather than $p(g)$. We train Discriminator (D) to maximise the probability of assigning the correct label to both training examples and samples from G . Simultaneously we train G to minimise $\log(1 - D(G(z)))$:

So in other words, D and G play the following two-player minimax game with value function $V(G, D)$:

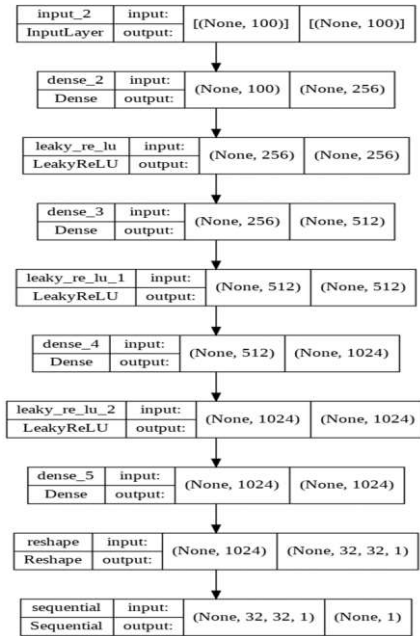
$$\min_G \max_D V(G, D) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (9)$$

Kang Zhang et al. [22] proposed a novel architecture of GAN with Multi-Layer Perceptron (MLP) as the discriminator and LSTM as the generator for forecasting the closing stock prices. The generator is built by LSTM to mine the data distribution of stocks from given data and generate data in the same distributions, whereas the discriminator designed MLP to discriminate between real and fake data. Their results showed that the novel GAN structure gave promising results compared with other models in machine learning and deep learning. HungChun Lin et al., [11] proposed stock price prediction model using GAN with Gated Recurrent Units (GRU) used as a generator that inputs historical stock price and generates future stock price and Convolutional Neural Network (CNN) as a discriminator to discriminate between real and fake distributions. Different to one step ahead forecasts only, the authors used deep learning to make multi-step ahead predictions more accurately. The authors also found that Wasserstein GAN performed better during unexpected events like COVID whereas conventional GAN performed better during normal times. They also showed that including RNN into a GAN makes it unstable because it is challenging to tune hyperparameters and without suitable parameters the results are poor. The key is tune parameters in each layer to make the whole model more accurate. Other researchers use Rainbow method based on Q-learning for hyperparameter optimisation. Martin Erdman et al., [7] used adversarial network with the Wasserstein distance to generate simulated detector data. The authors investigated two variants of GANs for detector simulations. In both cases, the transfer of probability distributions from one data set to another using GAN training worked well using the Wasserstein distance in the loss function. Instead of training a deep network with simulations that differ in detail from data, simulations can be adapted to match data prior to network training. With this method, authors showed that results are better compared to training with the originally simulated traces.

In this paper, we will use normalised log scale return VIX as our train data and normalised log scale return VVIX as our validation data.



We will then build ARIMA on scaled data and predict the fitted values on validation data. We also calculate the future trend growth with ARIMA. We will then make a dataframe of the fitted values and split it into train and test sets for our GAN model. We will add CNN conv1 layer as our discriminator and choose accuracy as our metrics. Then in generator we use two layers of CNN and LSTM together and again we choose accuracy as our metrics. For training the model, we compile discriminator and generator together and use the learning rate of 0.001 and beta of 0.5. A plot of the model is also created, and we can see that the model expects a 100-element point in laten space as input and will predict a single output.



A. RESULTS:

We ran the model for a total of 1200 epochs and it gave us the accuracy rate of 93.3%. We then use the fitted (predicted) values of GAN as features/inputs in SVM and RF. We define the RF model and use train features for training and fitting and test features for predicting. The results for MAPE and MAE are quite encouraging. For SVM, we use grid search and fit the model to get a score of 98.5% on fitting.

RF

MAPE: 0.0074763204010184145

MAE: 0.005130449308450033

SVM

MAPE: 0.005861350065975159

MAE: 0.004021246354831941

VII. CONCLUSIONS

The volatility of financial markets is a very pertinent research topic. It has been in the spotlight in the last 10 years due to Great Financial crisis, Covid and recent geo-political events. Many articles have been written modelling and forecasting volatility in various markets. Despite being a very popular topic, the literature has not drawn any conclusions as

to the most robust forecasting and computationally efficient model. A variety of different models have been proposed over the years, from the most popular such as ARIMA family of models and naïve models, such as Random Walk and Implied volatility models, to the Engel's autoregressive conditional variance (ARCH) family of models.

In the ongoing debate over finding the best volatility forecasting model, studies have shown to be in favour of the simple models. Central Banks pay very close attention to volatility and volatility of volatility as a factor of how loose or tight the financial conditions are. Hence, it has direct impact on things that most people care about such as their mortgage rates, inflation and unemployment.

Economics exhibit inter-dependencies, hesitation and reflexivity to this extent IFS based unsupervised techniques for data analysis seem to have an advantage.

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